



32-bit microcontrollers

HC32L072_HC32L073_HC32F072 series

IIS module

This product supports the following chip series

L Series	HC32L072	HC32L073
F Series	HC32F072	

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1 Summary

This application note introduces the I2S module in the HC32L072 / HC32L073 / HC32F072 series.

This application note includes:

- Introduction to the I2S module
- Introduction to audio formats
- Audio playback examples

Note:

- *This application note is a supplementary material for the application of the I2S module in the HC32L072 / HC32L073 / HC32F072 series and cannot replace the user manual. Please refer to the user manual for specific functions, register operations and other related matters.*

2 Introduction to I2S module

I2S (Inter-IC Sound) is an audio data transmission standard that uses a three-wire bus for serial and synchronous data transmission. Data is transmitted on the SD (serial data) line in little-endian format (MSB first), and the data length is usually 16/24/32 bits. Transmitter data is synchronized by the rising or falling edge of SCK (serial clock), and the receiver uses the falling edge of SCK.

SD Serial Data (mapped to the MOSI pin), used to send or receive data on two time-division multiplexed data channels (half-duplex mode only).

WS Word Select (mapped to the NSS pin), is the data control signal output in master mode and the data control signal input in slave mode.

CK Serial Clock (mapped to the SCK pin), is the serial clock output in master mode and the serial clock input in slave mode.

MCK master clock output (MCK is mapped to MISO pin). When I2S is configured in master mode (and MCKOE bit in I2Sx_PR register is set to 1), this additional clock is output with master clock (separate mapping). This clock is generated at a preconfigured frequency of $256 \times f_s$, where f_s is the audio signal sampling frequency.

Each I2S in this series supports master mode and slave mode, supports half-duplex transmission and reception, and if full-duplex mode is required, two I2S are required to cooperate.

The I2S peripheral supports four audio protocols (configurable by software):

- I2S Philips protocol
- MSB protocol
- LSB protocol
- PCM protocol (including PCM short frame and PCM long frame)

It also supports most audio frequencies (8 kHz, 16 kHz, 32 kHz, 44.1 kHz, 48 kHz, etc.).

The data format is programmable to 16, 24, or 32 bits of data length (per channel), MSB first, and 16 or 32 bits in length (per channel).

3 Introduction to audio format

In practical applications, most audio files are in MP3 format. MP3 is an audio compression technology, and its full name is Moving Picture Experts Group Audio Layer III, or MP3 for short. It is designed to significantly reduce the amount of audio data. Using the technology of MPEG Audio Layer 3, music is compressed into a smaller file at a compression rate of 1:10 or even 1:12, and for most users, the quality of the replayed sound is not significantly reduced compared to the original uncompressed audio. Audacity is a free audio processing software that follows the GNU protocol and can convert MP3 format to WAV format. WAV is one of the most common sound file formats. It is a standard digital audio file developed by Microsoft specifically for Windows. The file can record various mono or stereo sound information and ensure that the sound is not distorted. The format of the data itself is PCM or compressed, which is a lossless music format.

WAVE file is a very simple RIFF file, and its "format identifier" is defined as WAVE. RIFF chunk includes two sub-chunks, with IDs fmt and data, and an optional fact chunk. Fmt chunk is used to indicate the properties of audio data, including encoding method, number of channels, sampling frequency, number of bits required for each sample, etc. Fact chunk is an optional chunk, which is usually included when WAVE files are converted by some software. Data chunk contains the digitized waveform sound data of WAVE files. The overall structure of WAVE is shown in Figure 1.

Identifier ("RIFF")
Data size
Format identifier ("WAVE")
"fmt"
"fmt" block data size
"fmt" data
"fact" (optional)
"fact" block data size
"fact" data
"data"
Sound data size
Sound data

Figure 1 WAVE overall structure diagram

When the actual number of quantization bits for sound data is 16 bits, the data of different channels is arranged in the data chunk format.

Mono	Sample 1		Sample 2		
	Low Byte	High Byte	Low Byte	High Byte	
Dual Channel	Sample 1				
	Left Channel		Right Channel		
	Low Byte	High Byte	Low Byte	High Byte	

Figure 2 16-bit sound data format

4 Music playback example

Taking *Yang Shuo's Childhood.wav* file as an example, we can get the binary data of the WAV file through Visual Studio Code. We can get relevant information based on the file format.

1	Offset: 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F	
2	00000000: 52 49 46 46 76 E4 18 00 57 41 56 45 66 6D 74 20	RIFFvd..WAVEfmt.
3	00000010: 10 00 00 00 01 00 02 00 11 2B 00 00 44 AC 00 00	+...D,..
4	00000020: 04 00 10 00 64 61 74 61 E8 E3 18 00 13 1F AF 22	datahc..../"
5	00000030: 0A 3A E2 3A 05 25 12 27 B6 1C AF 1F 42 13 EA 0C	..b:.%.'6./..B.j.
6	00000040: 9F 0C 79 FC DE 0B 46 01 53 FF EF FD 95 EE 04 EF	..y ^..F..S.o}.n.o
7	00000050: 77 E1 7B DB 95 DC 59 D3 F3 E4 76 DF 10 EB E4 F0	wa{[.\YsSdv_.kdp
8	00000060: E7 F3 5A FF 6F 02 02 10 A5 0E 6D 19 1E 17 ED 22	gsZ.o...%.m...m"
9	00000070: B8 19 50 27 77 17 B9 1A 68 02 A0 01 7D EF E4 E2	8.P'w.9.h...}odb
10	00000080: 3B E0 D9 CD 91 D0 90 CC 71 C5 CE C8 39 C5 7C CB	;`YM.P.LqENH9E K
11	00000090: EB DC 58 E2 CE EF 89 FD 42 10 03 26 6B 2B ED 32	k\XbNo.}B..&k+m2
12	000000a0: 0D 31 4C 39 71 33 59 39 64 25 37 26 BA 14 8D 17	.1L9q3Y9d%7&:...
13	000000b0: 63 14 AB 18 9D 09 18 0E 68 00 0D 04 23 04 27 05	c.+.....h...#.'.
14	000000c0: FC F9 54 F4 29 F7 01 EC 67 ED D9 E4 10 DE E2 E0	yTt)w.lgmYd.^b`
15	000000d0: D9 E4 6A E5 5E F1 8D F0 A7 00 99 FC 56 1A 33 11	Ydje^q.p'.. V.3.
16	000000e0: 6E 20 FA 19 BE 22 39 1A C0 20 06 21 13 13 95 19	n.z.>"9.@...!....
17	000000f0: 15 FF FB 0B 71 EA F2 F7 58 D7 24 D6 9B CA EC C2	..{.qjrwXW\$V.JlB
18	00000100: 8C C4 B2 BE 49 C8 00 C9 5B E4 5B E5 78 FD DC FE	.D2>IH.I[d[ex]\~

Figure 3 wav file data

The ID of the WAV file is "RIFF", the size of the WAV file is 1631350, the type of the WAV file is "WAVE", the encoding format: 1 is PCM, the number of channels: 2 is dual-channel, the sampling frequency: 11025, the number of bytes per second: 44100, the number of words per sampling point: 4, the number of quantization bits: 16, and the amount of sound data is 1631208.

From the WAV file, it can be seen that the data starting from 0x2C is sound data. Taking 16-bit dual-channel data as an example, the data input to the I2S module is as follows:

0x1F13,0x22AF,0x3A0A,0x3AE2,0x2505,0x2712,0x1CB6,

Among them, 0x1F13 and 0x3A0A are input to the left channel in turn, and 0x22AF and 0x3AE2 are input to the right channel in turn. The sound frequency is configured to 44.1KHz.

5 Module configuration

The WM8731 decoding chip on the STK board sets the corresponding registers through the I2C interface and transmits the sound data in real time through the I2S interface, so that the headphones can hear the corresponding sound. I2S data transmission can be achieved through both interrupt mode and DMA mode.

The I2S module configuration process of this series of MCUs to play audio data is as follows:

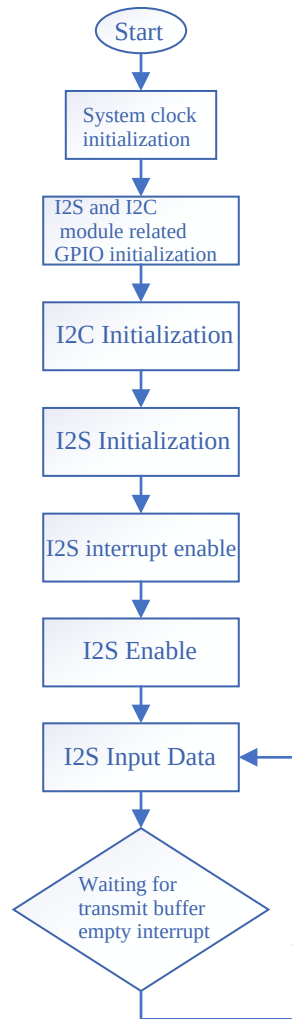


Figure 4 I2S module plays audio data

6 Reference examples and drivers

Through the above introduction, combined with the user manual of the HC32L072 / HC32L073 / HC32F072 series, we have a further understanding of the I2S module functions and operation methods of the above series of MCUs.

XHSC also officially provides application samples and driver libraries for this module. Users can open the sample project to further familiarize themselves with the application of this module and driver library. In actual development, they can also directly refer to the sample and use the driver library to quickly implement the operation of this module.

- Sample reference: ~/HC32_DDL/example/I2S
- Driver library reference: ~/HC32_DDL/driver/.../I2S

7 Conclusions

The above chapters briefly introduce the I2S module of the HC32L072 / HC32L073 / HC32F072 series, and explain in detail the various functions and operation steps of this module. In the actual application development process, if the user needs to further understand the usage and operation of this module, the corresponding user manual should prevail. The samples and driver libraries mentioned in this chapter can be used as further experiments and learning for users, and can also be directly applied in actual development.

8 Other information

Technical support information:

<http://www.xhsc.com.cn>

9 Version information & contact information

Date	Version	Revision content
2021/1/14	Rev1.0	Initial edition released.
2022/7/15	Rev1.1	Company Logo updated.



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